The Fairest in the Land?

Verbatim Transcriptions and Paralinguistic Features in the Mirror of Parliament Corpus

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LEVERHULME TRUST

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The Mirror of Parliament



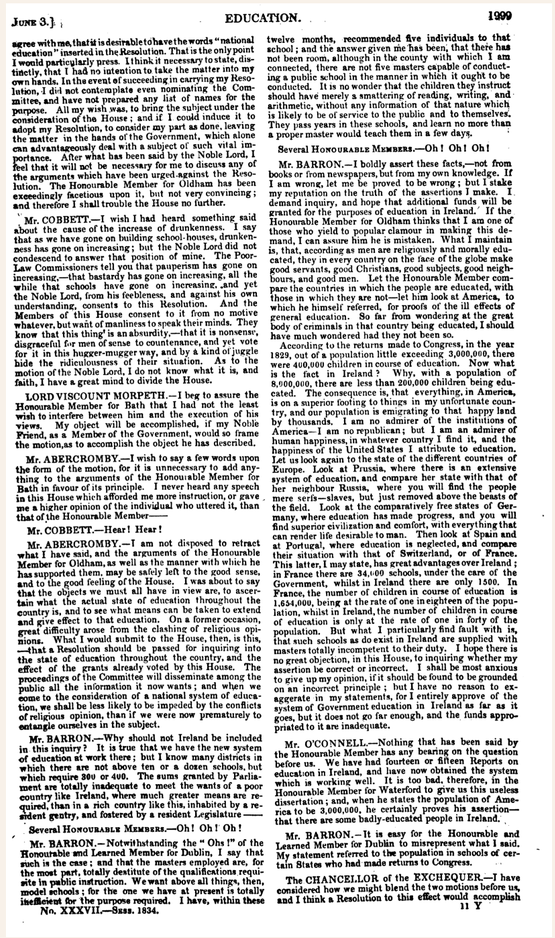
- Founded January 1828 by John Henry Barrow, barrister and journalist
- Weekly publication of parliamentary debates, both Houses
- Employed dedicated shorthand writers in the chamber
- One of those reporters: the young Charles Dickens, ‘the best reporter in the gallery’ (Knight 1865)

What Barrow Aimed For

‘…an attempt was boldly made by a gentleman named Barrow, to produce a verbal report of the proceedings of Parliament. He succeeded and carried it on for several years; and, for those years, I do not hesitate to say, that Barrow’s *Mirror of Parliament* is the primary record, and not *Hansard’s Debates*, because of the greater fulness which Barrow aimed at and obtained. It is within my own recollection – in the year 1833 or 1834 – just after the Reform Bill, that Gentlemen, who wanted to correct their speeches, did it for Barrow’s *Mirror of Parliament*.’

W. E. Gladstone, HC Deb 20 April 1877, c1576–77

What Does It Look Like?



JUNE 3.] EDUCATION. 1999

agree with me, that it is desirable to have the words "national education" inserted in the Resolution. That is the only point I would particularly press. I think it necessary to state, distinctly, that I had no intention to take the matter into my own hands. In the event of succeeding in carrying my Resolution, I did not contemplate even nominating the Committee, and have not prepared any list of names for the purpose. All my wish was, to bring the subject under the consideration of the House; and if I could induce it to adopt my Resolution, to consider my part as done, leaving the matter in the hands of the Government, which alone can advantageously deal with a subject of such vital importance. After what has been said by the Noble Lord, I feel that it will not be necessary for me to discuss any of the arguments which have been urged against the Resolution. The Honourable Member for Oldham has been exceedingly facetious upon it, but not very convincing; and therefore I shall trouble the House no further.

Mr. COBBETT.—I wish I had heard something said about the cause of the increase of drunkenness. I say that as we have gone on building school-houses, drunkenness has gone on increasing; but the Noble Lord did not condescend to answer that position of mine. The Poor-Law Commissioners tell you that pauperism has gone on increasing,—that bastardy has gone on increasing, all the while that schools have gone on increasing, and yet the Noble Lord, from his feebleness, and against his own understanding, consents to this Resolution. And the Members of this House consent to it from no motive whatever, but want of manliness to speak their minds. They know that this thing' is an absurdity,—that it is nonsensr, disgraceful for men of sense to countenance, and yet vote for it in this hugger-mugger way, and by a kind of juggle hide the ridiculousness of their situation. As to the motion of the Noble Lord, I do not know what it is, and faith, I have a great mind to divide the House.

LORD VISCOUNT MORPETH.—I beg to assure the Honourable Member for Bath that I had not the least wish to interfere between him and the execution of his views. My object will be accomplished, if my Noble Friend, as a Member of the Government, would so frame the motion, as to accomplish the object he has described.

Mr. ABERCROMBY.—I wish to say a few words upon the form of the motion, for it is unnecessary to add anything to the arguments of the Honourable Member for Bath in favour of its principle. I never heard any speech in this House which afforded me more instruction, or gave me a higher opinion of the individual who uttered it, than that of the Honourable Member——

Mr. COBBETT.—Hear! Hear!

Mr. ABERCROMBY.—I am not disposed to retract what I have said, and the arguments of the Honourable Member for Oldham, as well as the manner with which he has supported them, may be safely left to the good sense, and to the good feeling of the House. I was about to say that the objects we must all have in view are, to ascertain what the actual state of education throughout the country is, and to see what means can be taken to extend and give effect to that education. On a former occasion, great difficulty arose from the clashing of religious opinions. What I would submit to the House, then, is this, —that a Resolution should be passed for inquiring into the state of education throughout the country, and the effect of the grants already voted by this House. The proceedings of the Committee will disseminate among the public all the information it now wants; and when we come to the consideration of a national system of education, we shall be less likely to be impeded by the conflicts of religious opinion, than if we were now prematurely to entangle ourselves in the subject.

Mr. BARRON.—Why should not Ireland be included in this inquiry? It is true that we have the new system of education at work there; but I know many districts in which there are not above ten or a dozen schools, but which require 300 or 400. The sums granted by Parliament are totally inadequate to meet the wants of a poor country like Ireland, where much greater means are required, than in a rich country like this, inhabited by a resident gentry, and fostered by a resident Legislature——

Several HONOURABLE MEMBERS.—Oh! Oh! Oh!

Mr. BARRON.—Notwithstanding the "Ohs!" of the Honourable and Learned Member for Dublin, I say that such is the case; and that the masters employed are, for the most part, totally destitute of the qualifications requisite in public instruction. We want above all things, then, model schools; for the one we have at present is totally inefficient for the purpose required. I have, within these twelve months, recommended five individuals to that school; and the answer given me has been, that there has not been room, although in the county with which I am connected, there are not five masters capable of conducting a public school in the manner in which it ought to be conducted. It is no wonder that the children they instruct should have merely a smattering of reading, writing, and arithmetic, without any information of that nature which is likely to be of service to the public and to themselves. They pass years in these schools, and learn no more than a proper master would teach them in a few days.

Several HONOURABLE MEMBERS.—Oh! Oh! Oh!

Mr. BARRON.—I boldly assert these facts,—not from books or from newspapers, but from my own knowledge. If I am wrong, let me be proved to be wrong; but I stake my reputation on the truth of the assertions I make. I demand inquiry, and hope that additional funds will be granted for the purposes of education in Ireland. If the Honourable Member for Oldham thinks that I am one of those who yield to popular clamour in making this demand, I can assure him he is mistaken. What I maintain is, that, according as men are religiously and morally educated, they in every country on the face of the globe make good servants, good Christians, good subjects, good neighbours, and good men. Let the Honourable Member compare the countries in which the people are educated, with those in which they are not—let him look at America, to which he himself referred, for proofs of the ill effects of general education. So far from wondering at the great body of criminals in that country being educated, I should have much wondered had they not been so.

According to the returns made to Congress, in the year 1829, out of a population little exceeding 3,000,000, there were 400,000 children in course of education. Now what is the fact in Ireland? Why, with a population of 8,000,000, there are less than 200,000 children being educated. The consequence is, that everything, in America, is on a superior footing to things in my unfortunate country, and our population is emigrating to that happy land by thousands. I am no admirer of the institutions of America—I am no republican; but I am an admirer of human happiness, in whatever country I find it, and the happiness of the United States I attribute to education. Let us look again to the state of the different countries of Europe. Look at Prussia, where there is an extensive system of education, and compare her state with that of her neighbour Russia, where you will find the people mere serfs—slaves, but just removed above the beasts of the field. Look at the comparatively free states of Germany, where education has made progress, and you will find superior civilization and comfort, with everything that can render life desirable to man. Then look at Spain and at Portugal, where education is neglected, and compare their situation with that of Switzerland, or of France. This latter, I may state, has great advantages over Ireland; in France there are 34,000 schools, under the care of the Government, whilst in Ireland there are only 1500. In France, the number of children in course of education is 1,634,000, being at the rate of one in eighteen of the population, whilst in Ireland, the number of children in course of education is only at the rate of one in forty of the population. But what I particularly find fault with is, that such schools as do exist in Ireland are supplied with masters totally incompetent to their duty. I hope there is no great objection, in this House, to inquiring whether my assertion be correct or incorrect. I shall be most anxious to give up my opinion, if it should be found to be grounded on an incorrect principle; but I have no reason to exaggerate in my statements, for I entirely approve of the system of Government education in Ireland as far as it goes, but it does not go far enough, and the funds appropriated to it are inadequate.

Mr. O'CONNELL.—Nothing that has been said by the Honourable Member has any bearing on the question before us. We have had fourteen or fifteen Reports on education in Ireland, and have now obtained the system which is working well. It is too bad, therefore, in the Honourable Member for Waterford to give us this useless dissertation; and, when he states the population of America to be 3,000,000, he certainly proves his assertion—that there are some badly-educated people in Ireland.

Mr. BARRON.—It is easy for the Honourable and Learned Member for Dublin to misrepresent what I said. My statement referred to the population in schools of certain States who had made returns to Congress.

The CHANCELLOR of the EXCHEQUER.—I have considered how we might blend the two motions before us, and I think a Resolution to this effect would accomplish

Mirror of Parliament, 3 June 1834

come to the consideration of a national system of education, we shall be less likely to be impeded by the conflicts of religious opinion, than if we were now prematurely to entangle ourselves in the subject.

Mr. BARRON.—Why should not Ireland be included in this inquiry? It is true that we have the new system of education at work there; but I know many districts in which there are not above ten or a dozen schools, but which require 300 or 400. The sums granted by Parliament are totally inadequate to meet the wants of a poor country like Ireland, where much greater means are required, than in a rich country like this, inhabited by a resident gentry, and fostered by a resident Legislature ——

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Features of the Mirror

- Published weekly, so on average about three days after the event
- Paralinguistic markers in square brackets:
  - [Hear, hear!]
  - [Great laughter.]
  - [Cries of ‘No, no!’]
  - [With great warmth]
  - [Striking the Table with his hand]
- Audience reactions attributed to benches: *Opposition cheers*, *Ministerial laughter*

How Good Is It?

Jupp (1998) compared Mirror speech lengths against contemporaneous timings:

Speech	Reported duration	*Mirror* word count	Words/hour
Brougham on common law, 7 Feb 1828	6 hours 5 min	~36,000	~5,900
Hobhouse on Navarino, 14 Feb 1828	1 hour 45 min	~13,000	~7,400
Peel on Catholic Relief, 5 Mar 1829	4 hours 15 min	~24,000	~5,600
Lord Hawick, 8 Jun 1830	15 min	~1,400	~5,600

Jupp 1998: 236

~6,000 words an hour is the pace of a lecture carefully delivered from a text

Why Don't We Know About It?

- The *Mirror* was ruinously expensive to produce
- Barrow's personal loss: £5,000 (£450k/€520k in 2026 equivalent)
- Ceased publication October 1841
- No digital edition, survives as some scanned PDFs or hardcopies in research libraries
- Only a handful of scholarly citations (Brown 1955; McBath 1970; Jupp 1998)
- Barrow died in 1858, buried in a pauper's grave at Norwood

The Competition



— ‘I obtain them from the London newspapers, the country newspapers, and from the reports supplied by the Press Association. They are then passed into the hands of the collators…’ – T.C. Hansard Jr, 1888

— Often sent to speakers for correction or just used the speaker’s manuscript

— ‘…no more the speech uttered by them in the House of Commons than it is a Welsh ballad. […] We are duping the unborn generations. With open eyes we are sowing the seeds of dissension between historians of another age.’ (*Daily News*, 8 August 1853)

— ‘I hold myself bound for the *bona fides* of the reports, not for their literal accuracy’ – T.C. Hansard Jr, 1862

SIR EDWARD CODRINGTON.

Mirror of Parliament | 11-2_speech_001668 | 1834-02-25 | SMUGGLING ACT AMENDMENT BILL. | 235 tokens

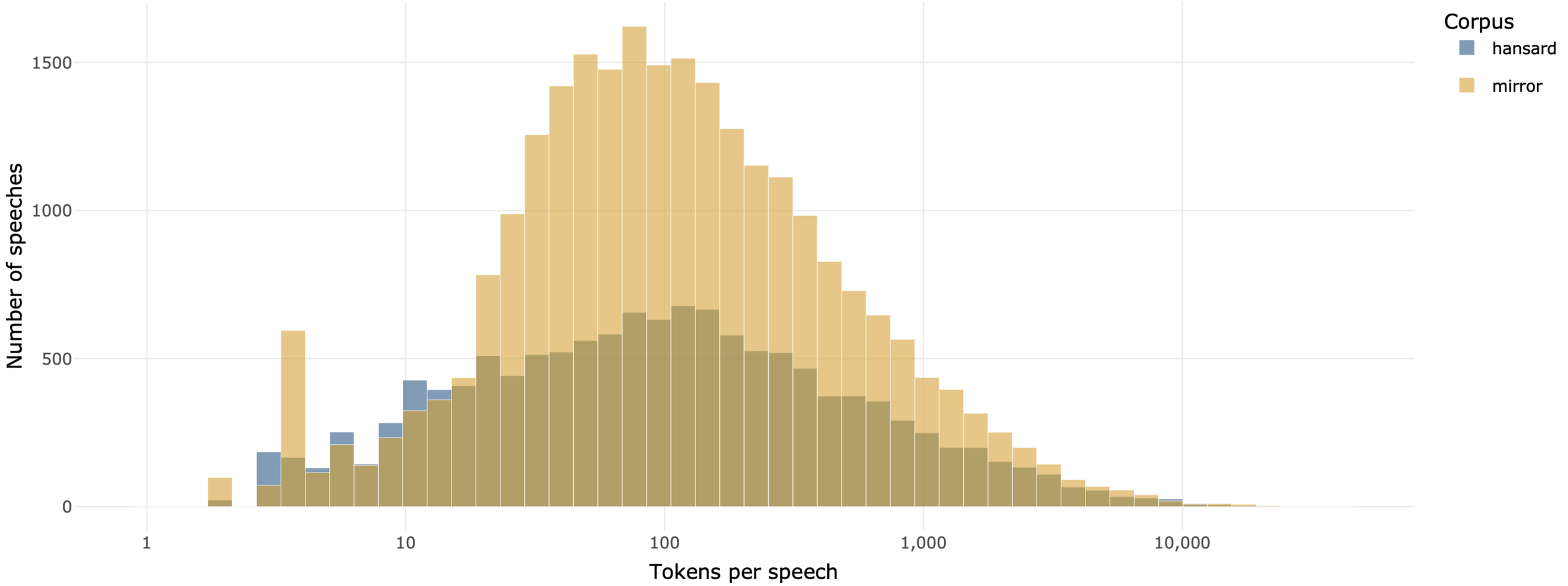
I shall be happy in giving my support to the Bill upon this one ground,—I object to anything which gives to a man-of-war the character of a prison. I cannot bear that a service, which ought to be as popular as it is honourable, should have anything in the shape of imprisonment associated with it. Upon a question of this kind, I cannot consider the interests of the smuggler so much as those of the service; and I maintain that it is a degradation to an admiral's flag that it should be flying in an absolute prison. Smugglers on board a King's ship are prisoners in every sense of the word. If they be sent abroad, the captain, with whom they sail, is responsible for their safe keeping; and if the period of service to which they are condemned be not expired on his return, he must transfer them to some other ship. In every sense of the word, therefore, these men are prisoners, and it is upon that ground that I take my stand. Nothing can induce me to think it proper to send men of this description on board his Majesty's ships. I will, therefore, give my cordial support to this Bill.

Sir Edward Codrington

Historic Hansard | hansard-1834_speech_000796 | 1834-02-25 | SMUGGLING ACT AMENDMENT BILL. | 129 tokens

said, that he could not, from his experience, concur with the hon Member. He should strongly object to making a man-of-war a common prison for felons. The naval service should not be made a punishment for crime. The great object should be to make it more comfortable than that of other employment; and was it the way to do that, to make our men-of-war a refuge for convicts? He did not wish to see the service thus degraded. It should be a voluntary service, and filled with men of good character; and that it could not be, if every flag ship were made a prison for convicts.

Mirror vs Hansard, same speech



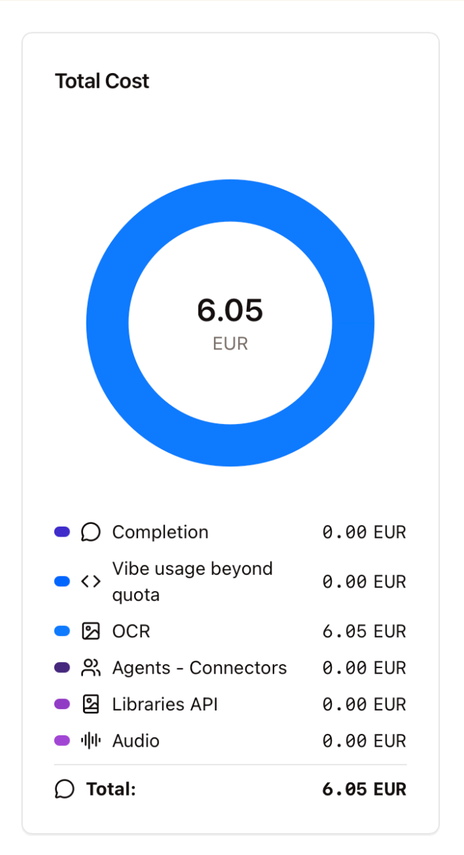
Distribution of speech lengths

203-Day Pilot Mirror Corpus

	Mirror of Parliament	*Hansard*
Speeches	25,432	12,936
Tokens	8,561,360	5,067,830
Unique speakers	2,969	1,355
Unique debate titles	2,596	835

- 1834: Tolpuddle Martyrs, Poor Law debates, Irish Church Appropriation, brought down Lord Grey's government
- 1839, 15 April to 3 June: Bedchamber Crisis, Melbourne's resignation, Peel's failed attempt to form a government, Victoria's refusal to change her ladies-in-waiting, Melbourne's return
- 1840, 16 January to 23 March: Prince Albert's precedence and income

Building the Pilot Mirror Corpus



- 6,300 pages of *Mirror* scans
- OCR via Mistral's vision language model API (mistral-ocr-2505, late November 2025)
- Total OCR cost for 8.5 million tokens of nineteenth-century parliamentary text: **€6.05**

Vision OCR for the Mirror

- VLM models (particularly Qwen and Mistral): text similarity scores up to 3-4 times higher than traditional OCR methods on complex scanned documents, especially on documents with complex layouts or poor scan quality
- Why Mistral? EU company, 94.9% accuracy across diverse document types, outperforming Google Document AI (83.4%) and Azure OCR (89.5%), n.b. 2025 figures(!)
  - Nineteenth-century parliamentary typography – longer-than-em-dashes, hyphenated line-breaks, running headers, column structure, small-caps speaker labels – handled well
  - Output: structured markdown per page
- Per-character confidence scores, standard in traditional OCR, are generally unavailable from these systems, complicating systematic quality evaluation
- In my manual checking of the Mirror transcriptions, 40 random pages, Mistral vastly outperformed Tesseract

Mistral vs Tesseract

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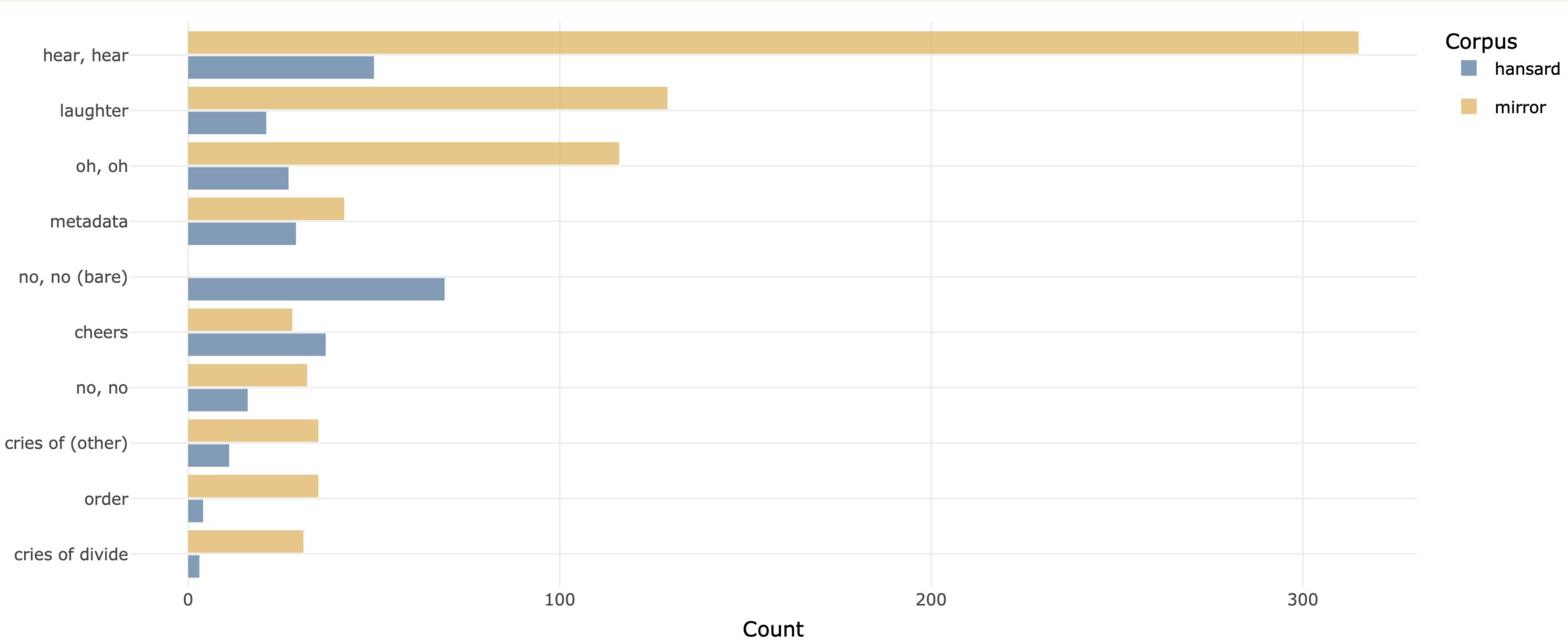
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Corpus Pipeline

- Raw OCR → correction pipeline (hyphen rejoin, spell-check with historical whitelist, LaTeX cleanup)
- Corrected pages → speech segmentation (speaker identification from period title conventions)
- Structured speeches → annotation (spaCy POS + lemma, PyMUSAS semantic tags)
- Parallel *Hansard* fetched from parliament.uk API into the same pipeline and schema (rather than refactor my existing corpus)
- Output: unified Parquet dataset, both corpora, ready for R/Python

Paralinguistics

- **939 paralinguistic markers** across the three pilot periods
- Classified into approval, disapproval, laughter, procedural, manner, interjection:



Paralinguistic markers by category

Marker categories

approval disapproval laughter
interjection procedural manner

Rank by

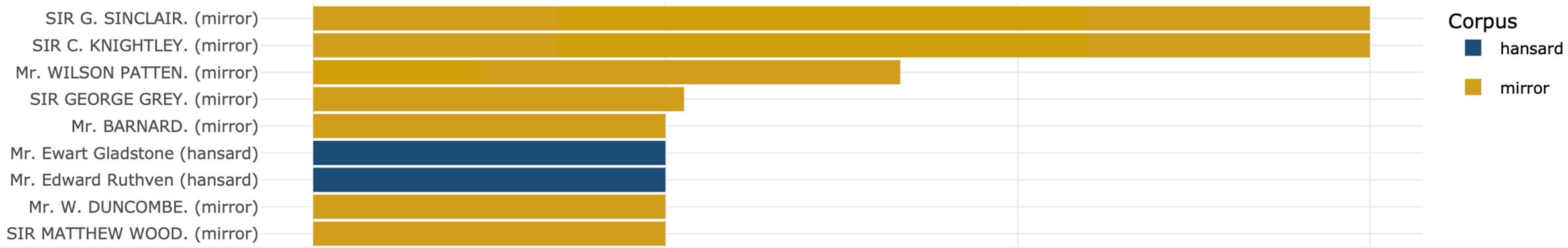
Absolute count
Per-speech rate

Min speeches per speaker

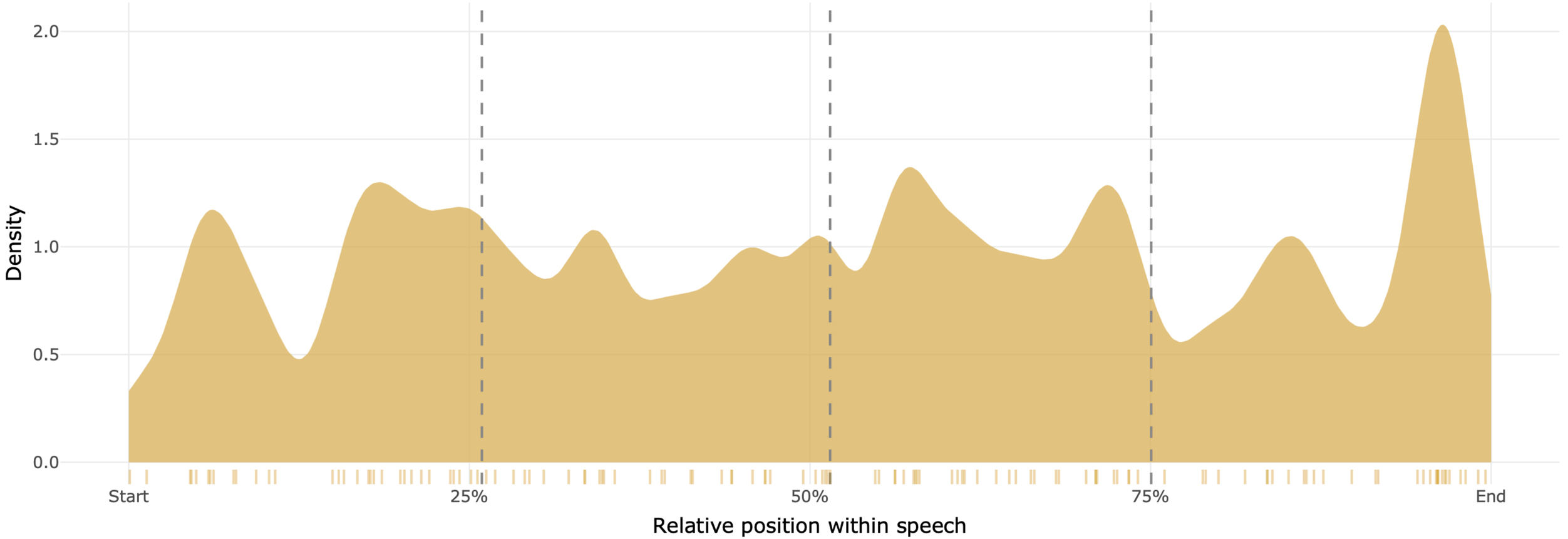
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Laughter markers



Where laughter falls

More and More Markers

	1834	1839	1840
Speeches	18,893	1,825	4,714
Markers	76	192	517
Per speech	0.004	0.105	0.110

- Marker density increases nearly 20× between 1834 and 1839
- The *Mirror* may be emphasising its verbatim-interpolation conventions?
  - Differentiation in the market?
  - Something the *Mirror's* shorthand reporters in the room can easily attest to systematically

Textual Reuse

- Of 175,468 candidate speech pairs, 1,885 have z-score ≥ 5 on longest common subsequence.
- Palmerston on the Constantinople-Egypt crisis, 27 March 1840:
  - *Hansard*: '**His** hon. Friend **thought** that the British Government and Lord Ponsonby, the British Ambassador at Constantinople, **had stimulated** the Sultan to renew hostilities against the Pacha of Egypt. **He (Lord Palmerston) could** assure him that he **was** entirely mistaken.'
  - *Mirror of Parliament*: '**My** honourable Friend **thinks** that the British Government and Lord Ponsonby, the British Ambassador at Constantinople, **stimulated** the Sultan to renew hostilities against the Pacha of Egypt. **I can assure him** that he **is** entirely mistaken.'
- '…one or two cases of *direct piracy* from the reports of the Mirror of Parliament, committed by a predecessor in this branch of literature, whose former pretensions should have withheld him…' J.H. Barrow, 6 December 1831

Comparative Signatures

- USAS semantic tag Z8m (he/him pronoun): **3.1× more frequent in *Hansard***
  - Third-person conversion at scale
  - *Hansard*: ‘The Hon. Member said *he* believed…’
  - *Mirror*: ‘*I* believe…’
- Grammatically:
  - *Mirror* has more INTJ (+22%), PRON (+3.8%)
  - *Hansard* has more PROPN (+4.9%)
  - The interjection and pronoun profile is a bit of a spontaneous-speech indicator
  - The proper noun excess in *Hansard*: third-person re-introducing speakers by name where *Mirror* leaves a pronoun

Conclusions

1. **OCR has changed:** Vision language models make digitisation of historical text dramatically cheaper; the barrier to building decent, quick corpora from nineteenth-century print has dropped by orders of magnitude

2. **We have good verbatim speech from this period:** Not full of hesitation markers or phonetic detail, but near-complete ‘tidied’ transcription of what was *generally said,* confirmed by independent timing evidence and with some paralinguistic markers showing where the audience reacted, when they laughed, when they objected

3. **The theatrics of Parliament:** Knowing where applause, laughter, and cries of ‘No, no!’ fell in a debate lets us see more of the performative culture of the 1830s Commons and Lords – and enables direct comparison with modern parliamentary behaviour

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Slides:

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SPARE: VISION LANGUAGE MODEL OCR

- Traditional OCR (Tesseract): character recognition → text
- VLM OCR (Mistral, GPT-4V) uses a unified vision-language model
- Divides the page image into small patches (15x15 pixels), embeds them with positional information, processes them through self-attention layers to produce a sequence of visual token embeddings
- Visual tokens projected into the same vector space as the language model's text tokens; transcription generated autoregressively, token by token, doing layout analysis, character recognition, and linguistic context in a single pass
- Significant quality gains on historical text (robust to degradation, diverse typefaces, and unusual layouts) combined with linguistic priors
- Central risk: system fails by producing fluent confabulation rather than transparent garbling

Spare: What Next

- Full 1828–1841 corpus (all surviving Mirror volumes)
- Systematic comparison with newspaper sources (*Times*, *Morning Chronicle*)
- Analysis of editorial overlap and plagiarism